

Transition Metal Engineering of Molybdenum Disulfide Nanozyme for Biomimicking Anti-Biofouling in Seawater

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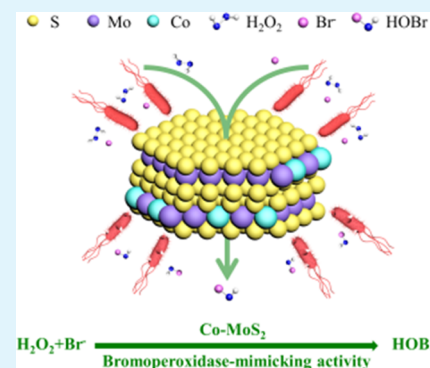
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Supporting Information

ABSTRACT: Nature has evolved diverse strategies to battle surface biofouling colonization and thus provides us novel insights into designing and developing advanced nontoxic antibiofouling materials and technologies. Mimicking the defense mechanisms of natural haloperoxidases in marine algae in response to biofilm colonization, here we show that the less active MoS₂ shows efficient haloperoxidase-mimicking activity through judicious transition metal engineering. Cobalt-doped MoS₂ (Co–MoS₂) displays an excellent haloperoxidase-mimicking performance in catalyzing the Br[−] oxidation into germicidal HOBr, roughly 2 and 23 times higher than the nickel-doped MoS₂ and pristine MoS₂, respectively. Accordingly, Co–MoS₂ shows an outstanding antimicrobial effect against drug-resistant bacteria and antibiofouling performance in real field tests in marine environments. The realization of robust haloperoxidase-mimicking activity of MoS₂ via metal engineering may open a new avenue to design highly active transition metal dichalcogenides for antibacterial and antibiofouling applications.

KEYWORDS: haloperoxidase, transition metal dichalcogenide, nanozyme, seawater, biofouling



INTRODUCTION

Mimicking the functionality of nature is an advanced solution to solve the critical challenges in technologies and applications.^{1–3} In nature, vanadium-dependent haloperoxidases as an essential protein in most marine algae,^{4–6} such as seaweed and some lichens, have been shown to play a critical role in response to biofouling by catalyzing the generation of biocidal hypohalous acid HOX (X: I[−], Br[−], or Cl[−]) from halide oxidation of halide ions with hydrogen peroxide (H₂O₂).^{7,8} For example, vanadium bromoperoxidases isolated from marine macroalgae *Delisea pulchra* utilize H₂O₂ for electrophilic Br[−] oxidation to generate HOBr.^{9,10} This natural defense mechanism represents a promising and environmentally friendly strategy to rival the current dominating antibiofouling solutions based on releasing toxic heavy metal biocides.^{11–16} However, the extraction difficulty, instability, and harsh reaction condition hinder their large-scale applications of natural enzymes.^{17–19}

Nanozymes, known as nanomaterials possessing enzyme-like functionalities, have been widely applied in biological systems because of their benign biocompatibility, chemical stability, tunable activity, and scalable manufacture capability.^{20–24} Recently, exciting paradigms can be found for haloperoxidase-mimicking nanozymes, such as CeO₂, V₂O₅, and tungsten engineering of metal–organic frameworks.^{25–27} However, even though some impressive preliminary results have been achieved, very few nanozymes with intrinsic haloperoxidase-like activity have been developed so far.

Layered transition metal dichalcogenides have demonstrated superior performance in many fields, such as solid lubricants, energy devices, petrochemical catalysis, and so forth, due to their physicochemical diversity, low cost, and chemical stability.^{28–33} In view of these advantages, MoS₂ has been explored as a valuable nanozyme that exhibits multiple enzyme-mimicking activities, such as superoxide dismutases, catalases, and peroxidases, in combating oxidative stress and disinfecting drug-resistant bacteria.^{34,35} Unfortunately, MoS₂ exhibited very poor haloperoxidase-mimicking activity, as experimentally revealed in this work. In fact, for a variety of catalytic applications, experimental and computational results have confirmed that the activity origin of layered MoS₂ could be attributed to the highly exposed edge sites.^{36–38} From this perspective, significant research efforts are being conducted to regulate the catalytic performance of MoS₂, including dimensional control, metallic and non-metallic doping, and defect engineering.^{38–44} Among them, heteroatom engineering is an appealing pathway to enhance the redox ability of MoS₂.^{45,46} We therefore anticipate that the metallic heteroatom-doped MoS₂ could enhance haloperoxidase-mimicking activity and

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enrich our understanding on the relationship between the foreign metallic doping and biological mimicking reaction kinetically.

Herein, we show that less active MoS₂ engineered with some transition metals exhibited intrinsic haloperoxidase-mimicking activity for inhibiting surface build-up of biofouling (Figure 1).

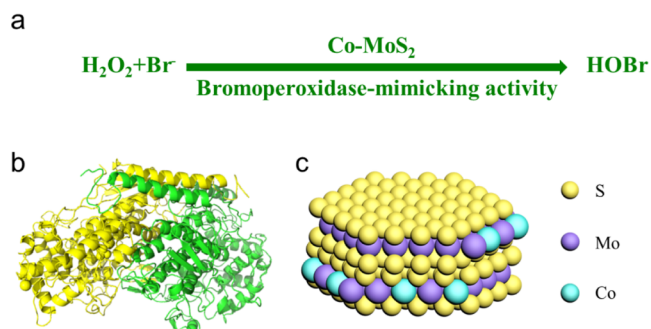


Figure 1. (a) Reaction scheme of Br[−] oxidation by bromoperoxidase-mimicking Co–MoS₂ in the presence of H₂O₂. (b) Natural vanadium bromoperoxidase (RCSB PDB, 1QHB). (c) Schematic illustration of the Co–MoS₂ nanosheet.

Our results reveal that cobalt-doped MoS₂ nanosheet (hereafter, referred to as Co–MoS₂) yielded superior haloperoxidase-mimicking capacity toward the catalytic oxidation of Br[−] with H₂O₂. The as-prepared Co-doped MoS₂ showed exceptional broad-spectrum antibacterial capability toward *E. coli* (90%) and *S. aureus* (88%). The anti-biofouling capability of Co-doped MoS₂ was favorably illustrated in marine environments when used as surface coating additives. This work not only sheds light on modulating haloperoxidase-mimicking properties of MoS₂ via element engineering approach but also paves an avenue for broadening the application ranges of transition metal dichalcogenides.

RESULTS AND DISCUSSION

Co–MoS₂ nanosheet was obtained through a solvothermal method (see the Experimental Section). The X-ray diffraction (XRD) peaks centered at 13.8, 33.5, 39.7, and 59.3° were typically assigned to the (002), (100), (103), and (110) crystal planes of 2H–MoS₂, respectively (Figure S1).⁴⁷ We also prepared MoS₂ engineered with other transition metals (Fe, Ni, Cu, and Zn) under the similar preparation conditions. No obvious difference was observed in the XRD patterns, revealing that no new phases were produced after transition metal doping.⁴⁸ Co–MoS₂ and the pristine MoS₂ possessed a two-dimensional nanosheet morphology with a sheet size of 1–2

μm, as observed from the scanning electron microscopy (SEM) image (Figures 2a and S2). Also, no large nanoparticles or clusters related to the doping were found. High-resolution transmission electron microscopy (HRTEM) observation suggested that Co–MoS₂ exhibited dense interconnected corrugations, which is comparable with the pristine MoS₂ (Figure S3). The interlayer lattice fringe spacing of as-prepared Co–MoS₂ was roughly estimated to be 0.65 nm. Energy-dispersive X-ray mapping results confirmed that the Co element was homogeneously distributed in Co–MoS₂ (Figure 2c). The full X-ray photoelectron spectroscopy (XPS) is shown in Figure S4. The high-resolution S 2p core level spectrum displayed two obvious peaks located at 162.3 and 163.6 eV, which could be ascribed to the divalent sulfur ions (S^{2−}) (Figure 3a).^{49,50} An additional peak at 163.7 eV corresponding to the Co–S bond was also observed, indicating that Co was successfully incorporated into MoS₂. Moreover, Mo 3d peaks (232.4 and 229.3 eV) of Co–MoS₂ exhibited negative shift, revealing the occurrence of electron transfer from Mo to the doped Co element.⁵¹ This phenomenon could be also observed in other transition metal-engineered MoS₂ (Figures S5 and S6). The presence of two Co 2p peaks at 779.3 and 794.1 eV in Figure 3b directly verified the existence of Co species.⁵² The Co amount in Co–MoS₂ was determined to be about 2.5 wt %. These results indicated the covalent doping of Co elements into the lattice of the MoS₂.^{53–55} Two peaks located at 400.8 and 373.5 cm^{−1} were observed in the Raman spectra (Figure S7), which could be ascribed to A_{1g} and E_{2g} vibrations of MoS₂, respectively.⁵⁶ Notably, the positive shift of the E_{2g} suggested the lattice distortion during the Co doping. In comparison to pristine MoS₂ and Ni–MoS₂, Co–MoS₂ exhibited a significantly stronger electron paramagnetic resonance (EPR) signal at *g* = 2.004 (Figure 3c), illustrating that Co–MoS₂ had a higher S-vacancy concentration with the introduction of Co.^{57–60}

To test if Co–MoS₂ possesses haloperoxidase-mimicking activity in promoting the Br[−] oxidation with H₂O₂ into HOBr, we used the phenol red bromination array that involves the selective formation of bromophenol blue (λ = 590 nm).⁶¹ Such a bromination process would induce a visible color change from yellow to blue, which could be monitored spectrophotometrically. As revealed in Figure 4a, the intensity of characteristic absorbance at 590 nm significantly increased with the extension of reaction time (pH = 5.7), indicating the intrinsic haloperoxidase-mimicking capacity of Co–MoS₂. By contrast, the pristine MoS₂ exhibited a poor catalytic activity. We also examined the catalytic ability of other transition metal-doped MoS₂ under similar conditions. Experimental results demonstrated that Ni–MoS₂ also yielded obvious haloperox-

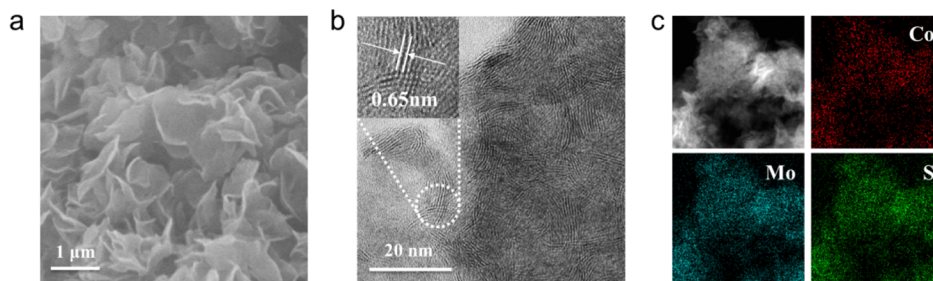


Figure 2. (a) SEM image of the layered Co–MoS₂ nanosheet. (b) HRTEM image of the Co–MoS₂ nanosheet. (c) EDS mapping analysis of Co–MoS₂.

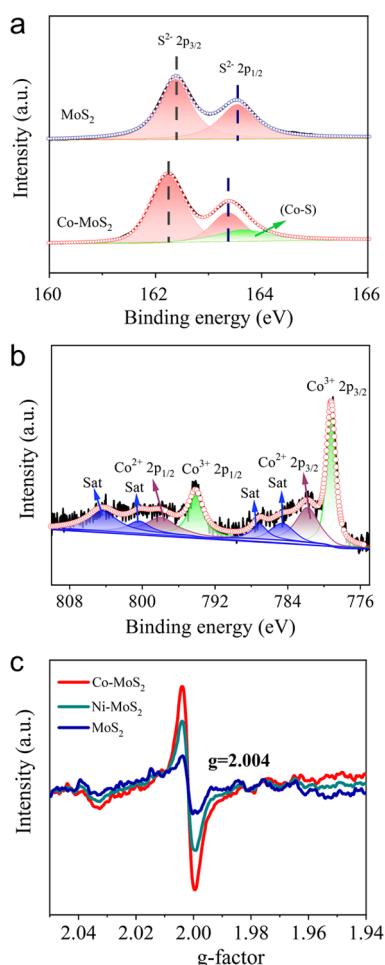


Figure 3. (a) S 2p XPS spectra of pristine MoS₂ and Co-MoS₂. (b) Co 2p XPS spectra of Co-MoS₂. (c) EPR spectra of pristine MoS₂, Co-MoS₂, and Ni-MoS₂.

idase-mimicking activity, but other metals (Zn, Fe, and Cu) did not promote the catalytic activity of MoS₂, as shown in Figure 4b. It is also found that Co-MoS₂ shows an excellent haloperoxidase-like activity at a concentration of 0.05 mg mL⁻¹, about 2- and 23-fold higher than Ni-MoS₂ and pristine MoS₂, respectively. The activity improvement should be ascribed to the higher sulfur vacancy concentration of Co-MoS₂ than that of MoS₂ and Ni-MoS₂, as evidenced by EPR analysis (Figure 3c). The catalytic activity of Co-MoS₂ was

also pH-dependent (Figure S8). A 52% activity at pH 8.0 was obtained in comparison with that at pH 5.7. The decreased performance at a high pH could be ascribed to several reasons.²⁵ The reactivity of Co-MoS₂, phenol red, and H₂O₂ decreased in the alkaline environments. In addition, the higher pH could result in the decreased stability of the HOBr and other oxidized bromine intermediates. The intermediate of bromination reaction of phenol red was examined. It was found that Co-MoS₂ with Br⁻ and H₂O₂ oxidant resulted in the gradually decayed celestine blue (Figure 4c), clearly suggesting the generation of HOBr intermediate because the celestine blue indicator would be decomposed by oxidized halide species.⁶²

The steady-state reaction kinetics of haloperoxidase-mimicking catalytic reaction were further evaluated. It was clearly observed that the initial bromination rate exhibited a proportional dependence on the mass concentration of Co-MoS₂ (Figure S9). The Br⁻ or H₂O₂ substrate showed a typical Michaelis–Menten behavior in the presence of other constant reactants (Figure 5a,b). For comparison, we also evaluated the kinetic behavior of Co-MoS₂ with different Co loadings by varying the initial cobalt precursor amount of 0.18, 0.27, and 0.36 mmol [denoted as Co-MoS₂-*x* (*x* = 1, 2, 3)]. As demonstrated, Co-MoS₂-2 showed the optimal catalytic performance. The Michaelis constant *K_m* values for Br⁻ and H₂O₂, determined by Lineweaver–Burk plots, are 28 mM and 117 μM, respectively. These values are significantly lower compared with that of MoS₂ doped with other transition metals (Fe, Ni, Cu, and Zn) (Figure S10 and Table S1). The lower *K_m* of H₂O₂ suggests its stronger affinity with Co-MoS₂.²⁵ Figure 5c shows that the haloperoxidase-mimicking ability of Co-MoS₂ nanozymes could remain stable over several catalytic cycles.

Bacterial accumulation to submerged surfaces is considered as the initial stage during biofouling and influences the subsequent settlement of micro-/macroorganisms.^{63,64} With the haloperoxidase-mimicking properties, we first examined the antibacterial capacity of Co-MoS₂ and MoS₂ nanozymes. Two typical bacteria, *E. coli* and *S. aureus*, were applied as the model bacteria and co-incubated with the synthesized nanozymes in the presence of H₂O₂ and Br⁻. To investigate whether the biocidal effect caused by Co-MoS₂ was through the haloperoxidase mimicking-mediated mechanism, we examined the antibacterial performance by investigating the absorption of the resulting bacterial suspension at 592 nm (OD₅₉₂) after co-incubation with different groups (Figure 6a,b). Co-MoS₂/

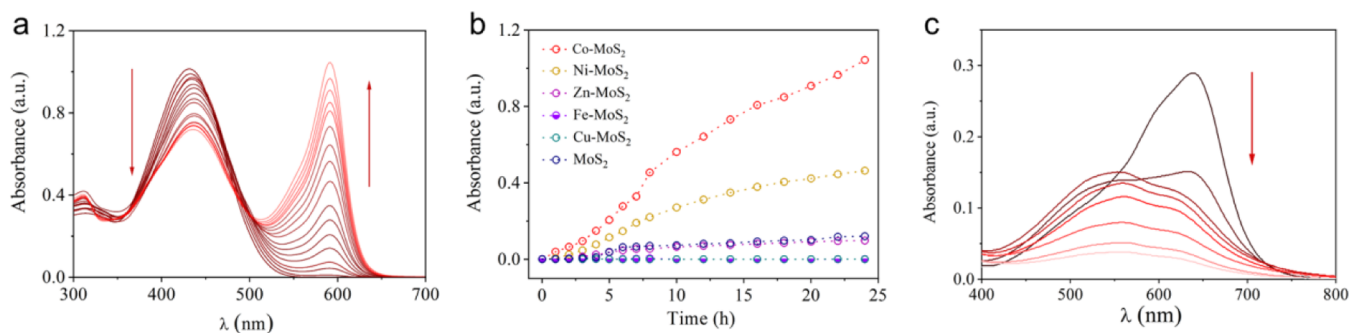


Figure 4. (a) Time-dependent optical absorption of phenol red solution with Co-MoS₂, H₂O₂, and Br⁻ at a pH value of 5.7. (b) Time-dependent absorbance variation of phenol red solution at 590 nm with the use of different transition metal-doped MoS₂ nanozymes. (c) Optical absorption spectra of celestine blue solution with H₂O₂, Br⁻, and Co-MoS₂.

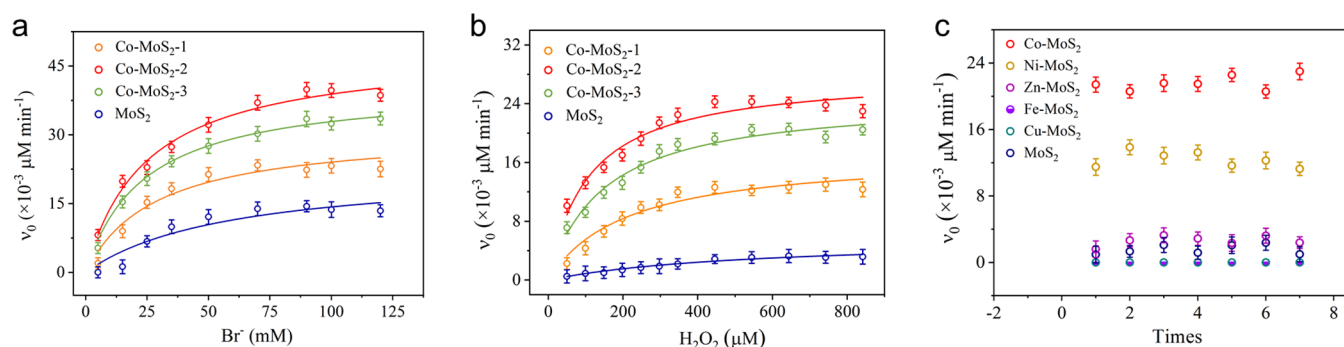


Figure 5. (a) Kinetic curve of the phenol red bromination as a function of H_2O_2 (0–900 μM); nanozyme: 0.05 mg mL^{-1} , Br^- : 25×10^{-3} M, phenol red: 5×10^{-5} M. (b) Kinetic curve of phenol red bromination as a function of Br^- substrate (0–125 mM); nanozyme: 0.05 mg mL^{-1} , H_2O_2 : 350×10^{-6} M, phenol red: 5×10^{-5} M. (c) Stability of transition metal-doped MoS₂ nanozymes toward the phenol red bromination.

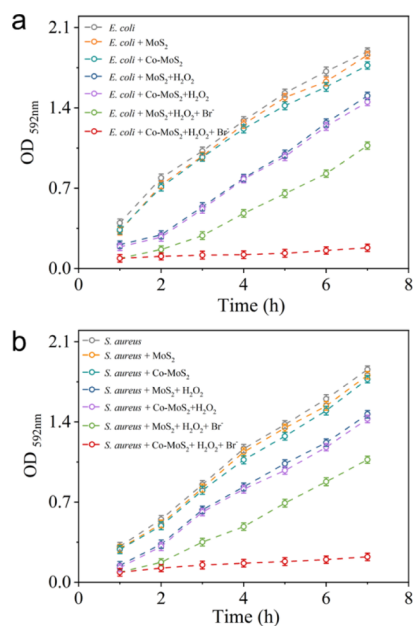


Figure 6. Bacterial growth curves of (a) *E. coli* and (b) *S. aureus* co-incubated with different additives, nanozyme: 1.0 mg mL^{-1} , H_2O_2 : 35 μM , Br^- : 25 mM.

MoS₂ alone or Co-MoS₂/MoS₂+Br⁻ exerted a negligible antibacterial activity. The Co-MoS₂/MoS₂+H₂O₂ control groups exhibited a disinfection efficiency of 18%/19% and 20%/18% for *E. coli* and *S. aureus*, respectively, which should

be due to the moderate antibacterial capacity of H_2O_2 . Notably, Co-MoS₂ showed the excellent antibacterial activity for bacterial strains in the presence of H_2O_2 and Br⁻, with a disinfection capacity of 90 and 88% for *E. coli* and *S. aureus*, respectively, which was significantly higher than MoS₂ (42% for *E. coli* and 40% for *S. aureus*) under the identical conditions. These results imply that the germicidal capacity of Co-MoS₂ in this study was originated from the haloperoxidase-mimicking catalytic mechanism. After a further incubation (37 °C, 12 h), the bacterial colonies with Co-MoS₂ + H₂O₂ + Br⁻ dramatically decreased compared to other reference groups (Figure 7).

The average concentration of Br⁻ in seawater is approximately 0.8 mM.⁶⁵ Also, H_2O_2 can be generated in natural seawater in a concentration between 0.1 and 0.25 μM .⁶⁶ We then examined the anti-biofouling performance of Co-MoS₂ and MoS₂ in natural marine environments. The real field evaluation was conducted at Haikou Bay (Haikou, China) by immersing the painted plates containing Co-MoS₂ or MoS₂ additives into seawater with an immersion depth of ~ 1 m for 45 days. The blank plates without any antibiofouling additives were fully covered with marine biofoulants, such as barnacles and sponges (Figure 8a). The reference panels with 5 wt % pristine MoS₂ were less fouled (Figure 8b). Notably, the plates with 5 wt % Co-MoS₂ displayed the best resistance to marine biofoulants in the testing site (Figure 8c). These results further confirm that the significant antibiofouling effect of Co-MoS₂ was originated from the haloperoxidase-mimicking activity.

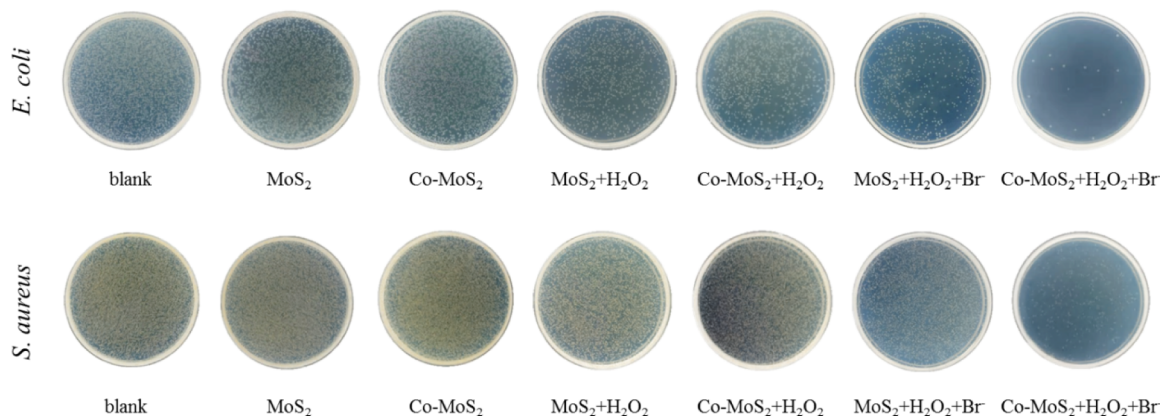


Figure 7. Digital images of the model bacteria after co-incubation with different combinations.

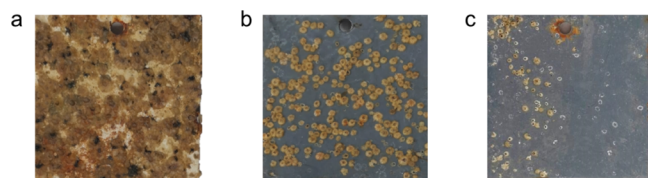


Figure 8. Representative digital images of painted panels with or without nanozyme additives: (a) blank, (b) pristine MoS₂, and (c) Co-MoS₂.

CONCLUSIONS

In conclusion, transition metals engineering the MoS₂ nanozyme have been synthesized and explored as functional analogues of natural haloperoxidases for antibiofouling for the first time. By doping with several transition metals, it was confirmed that Co-MoS₂ yielded 2 and 23 times higher haloperoxidase-mimicking activity (the catalytic conversion of Br[−] and H₂O₂ into the germicidal HOBr) than Ni-MoS₂ and MoS₂, respectively. This superior haloperoxidase-mimicking ability resulted in a significant antimicrobial effect against drug-resistant bacteria and biofilm colonization in real marine environments. This study paves the way for the potential application of layered transition metal dichalcogenides in combating biofouling.

EXPERIMENTAL SECTION

Reagents. Ammonium tetrathiomolybdate ((NH₄)₂MoS₄, 99.95%), cobalt nitrate hexahydrate (Co(NO₃)₂·6H₂O, 99%), phenol red (C₁₉H₁₄O₅S, ACS grade), zinc nitrate hexahydrate (Zn(NO₃)₂·6H₂O, 99%), ammonium bromide (NH₄Br, 99.998%), and copper nitrate trihydrate (Cu(NO₃)₂·3H₂O, 99.99%) were received from Aladdin Biochemical Technology Co., Ltd. H₂O₂ (30 wt.%), sodium chloride (NaCl, 99.5%), and nickel nitrate hexahydrate (Ni(NO₃)₂·6H₂O, 98%) were obtained from Xilong Chemical Co., Ltd. Iron nitrate nonahydrate (Fe(NO₃)₃·9H₂O, 99.99%) and *N,N*-dimethylformamide (C₃H₇NO) were obtained from Macklin Biochemical Co., Ltd. All these chemicals in this study were employed directly without any purification.

Synthesis of the MoS₂ and Transition Metal-Doped MoS₂. 0.27 mmol cobalt nitrate hexahydrate or other transition metal nitrate salts and 1.9 mmol ammonium tetrathiomolybdate were dissolved into 70 mL of *N,N*-dimethylformamide and subsequently sealed in a 100 mL Teflon-lined autoclave. After keeping at 190 °C for 24 h, the samples were collected, washed, and dried. Finally, the black samples were treated at 350 °C for 4 h in a nitrogen atmosphere. The preparation of other transition metal-doped MoS₂ was the same as that of Co-MoS₂. For the synthesis of pristine MoS₂, the procedure was similar with Co-MoS₂ but without the addition of metal nitrate salt.

Characterizations. XRD data were obtained by a Shimadzu XRD-6000 (λ = 1.54178 Å, Cu Kα). The morphology of the as-synthesized nanozymes was characterized by a Hitachi S-4800 scanning electron microscope (5 kV) and a JEOL-2010 transmission electron microscope (200 kV). X-ray photoelectron spectroscopy was conducted on an ESCALAB 250. The electron paramagnetic resonance technique was applied to evaluate the sulfur vacancy (EPR, Bruker, MS-5000). The optical absorption and kinetic analyses were performed on a UV-1800 PC spectrometer.

Catalytic activity Evaluation. 50 μg mL^{−1} nanozyme was dispersed in 100 mL of MilliQ-water containing 50 μM phenol red and 25 mM NH₄Br. Then, 350 μM H₂O₂ was injected into the abovementioned solutions to initiate the enzyme-mimicking reaction. The resulting solution were collected and filtered with a 0.45 μm PTFE filter. The absorption spectra were subsequently measured on a UV-vis spectrometer.

Kinetics assay. The kinetic evaluation of the nanozyme reaction was obtained by changing one agent, while holding other substrate constants. The concentration-dependent kinetic plots were obtained by tracing the optical absorbance of bromophenol blue at 590 nm.⁶¹ The concentration of bromophenol blue (extinction coefficient: 72,200 M^{−1} cm^{−1})²⁵ was characterized using Lambert–Beer law (eq 1). Finally, the kinetic plots were analyzed using the Lineweaver–Burk linearization method as shown below in eqs 2–4

$$\frac{1}{\nu} = \frac{K_m}{\nu_{\max}[C]} + \frac{1}{\nu_{\max}} \quad (1)$$

$$[Br_4PR] = \frac{A_{590}}{d \cdot \epsilon_{Br_4PR}} \quad (2)$$

$$\nu_0 = \frac{d[Br_4PR]}{dt} = \text{const.} + \frac{\nu_{\max} \bullet [S]}{K_m + [S]} \quad (3)$$

$$\frac{1}{\nu_0} = \frac{1}{\frac{d[Br_4PR]}{dt}} = \frac{K_m}{\nu_{\max}} \bullet \frac{1}{[S]} + \frac{1}{\nu_{\max}} \quad (4)$$

Antibacterial Test. Drug-resistance bacteria *E. coli* and *S. aureus* were selected in this study and subsequently cultured in the Luria–Bertani (LB) substrate (5 mL) at 37 °C on a horizontal shaker for 10 h. Then, 100 μL of the cultured bacteria solution was co-incubated with different combinations at 37 °C for a period of 3 h: (I) nanozyme, (II) nanozyme + H₂O₂, and (III) nanozyme + H₂O₂ + Br[−]. The time-dependent growth behavior of the bacteria solution was traced by optical absorption. Finally, the resulting bacteria were spread on agar-LB plates and further cultured at 37 °C for another 10 h.

Field test. To prove the practicability of the as-prepared Co-MoS₂ samples, the antibiofouling was evaluated in real marine environments (Haikou bay, Hainan, China). The stainless-steel plates (4 cm × 4 cm in size) were polished using abrasive sandpaper and washed with water and ethanol. 5 wt % nanozyme powder was added into the commercially available boat hull paint and homogenized by continuous ultrasound and stirring. The as-obtained paste was then coated on the stainless-steel substrates using brush. After drying, the coated plates were fixed on a steel framework and immersed in seawater at a depth of about 1 m.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.2c00172>.

XRD pattern and XPS spectra analysis of MoS₂ and transition metal-doped MoS₂, SEM and HRTEM images of MoS₂, Raman spectra of MoS₂ and Co-MoS₂, the effect of mass concentration of nanozymes and pH on the bromination activity, and kinetic analysis and their fitting (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Green, J. J.; Elisseeff, J. H. Mimicking Biological Functionality with Polymers for Biomedical Applications. *Nature* **2016**, *540*, 386–394.
- (2) Sedó, J.; Saiz-Poseu, J.; Busqué, F.; Ruiz-Molina, D. Catechol-based Biomimetic Functional Materials. *Adv. Mater.* **2013**, *25*, 653–701.
- (3) Jo, C. H.; Voronina, N.; Sun, Y. K.; Myung, S. T. Gifts from Nature: Bio-Inspired Materials for Rechargeable Secondary Batteries. *Adv. Mater.* **2021**, *33*, 2006019.
- (4) Wang, W.; Luo, Q.; Li, J.; Li, Y.; Wu, R.; Li, Y.; Huo, X.; Wang, N. Single-Atom Tungsten Engineering of MOFs with Biomimetic Antibiofilm Activity toward Robust Uranium Extraction from Seawater. *Chem. Eng. J.* **2022**, *431*, 133483.
- (5) Punitha, T.; Phang, S.-M.; Juan, J. C.; Beardall, J. Environmental Control of Vanadium Haloperoxidases and Halocarbon Emissions in Macroalgae. *Mar. Biotechnol.* **2018**, *20*, 282–303.
- (6) Wu, R.; Wang, W.; Luo, Q.; Zeng, X.; Li, J.; Li, Y.; Li, Y.; Li, J.; Wang, N. Room Temperature Synthesis of Defective Cerium Oxide for Efficient Marine Anti-Biofouling. *Adv. Compos. Hybrid Mater.* **2021**, DOI: 10.1007/s42114-021-00256-7.
- (7) Butler, A.; Sandy, M. Mechanistic Considerations of Halogenating Enzymes. *Nature* **2009**, *460*, 848–854.
- (8) Wever, R.; Tromp, M. G. M.; Krenn, B. E.; Marjani, A.; van Tol, M. Brominating Activity of the Seaweed *Ascophyllum nodosum*: Impact on the Biosphere. *Environ. Sci. Technol.* **1991**, *25*, 446–449.
- (9) de Boer, E.; Plat, H.; Tromp, M. G. M.; Wever, R.; Franssen, M. C. R.; van der Plas, H. C.; Meijer, E. M.; Schoemaker, H. E. Vanadium Containing Bromoperoxidase: An Example of An Oxidoreductase

with High Operational Stability in Aqueous and Organic Media. *Biotechnol. Bioeng.* **1987**, *30*, 607–610.

(10) Sandy, M.; Carter-Franklin, J. N.; Martin, J. D.; Butler, A. Vanadium Bromoperoxidase from *Delisea pulchra*: Enzyme-Catalyzed Formation of Bromofuranone and Attendant Disruption of Quorum Sensing. *Chem. Commun.* **2011**, *47*, 12086–12088.

(11) Hao, X.; Chen, S.; Qin, D.; Zhang, M.; Li, W.; Fan, J.; Wang, C.; Dong, M.; Zhang, J.; Cheng, F.; Guo, Z. Antifouling and Antibacterial Behaviors of Capsaicin-based pH Responsive Smart Coatings in Marine Environments. *Mater. Sci. Eng. C* **2019**, *108*, 110361.

(12) Liu, M.-J.; Li, P.; Meng, Q.-W.; Ge, Q. Membranes Constructed by Metal-Ligand Complexation for Efficient Phosphorus Removal and Fouling Resistance in forward Osmosis. *Adv. Compos. Hybrid Mater.* **2021**, DOI: 10.1007/s42114-021-00254-9.

(13) Mai, X.; Mai, J.; Liu, H.; Liu, Z.; Wang, R.; Wang, N.; Li, X.; Zhong, J.; Deng, Q.; Zhang, H. Advanced Bamboo Composite Materials with High-Efficiency and Long-Term Anti-Microbial Fouling Performance. *Adv. Compos. Hybrid Mater.* **2021**, DOI: 10.1007/s42114-021-00380-4.

(14) Hao, X.; Wang, W.; Yang, Z.; Yue, L.; Sun, H.; Wang, H.; Guo, Z.; Cheng, F.; Chen, S. pH Responsive Antifouling and Antibacterial Multilayer Films with Self-Healing Performance. *Chem. Eng. J.* **2018**, *365*, 130–141.

(15) Dobretsov, S.; Teplitski, M.; Paul, V. Mini-Review: Quorum Sensing in the Marine Environment and Its Relationship to Biofouling. *Biofouling* **2009**, *25*, 413–427.

(16) Kristensen, J. B.; Meyer, R. L.; Laursen, B. S.; Shipovskov, S.; Besenbacher, F.; Poulsen, C. H. Antifouling Enzymes and the Biochemistry of Marine Settlement. *Biotechnol. Adv.* **2008**, *26*, 471–481.

(17) Liang, M.; Yan, X. Nanozymes: from New Concepts, Mechanisms, and Standards to Application. *Acc. Chem. Res.* **2019**, *52*, 2190–2200.

(18) Gao, L.; Fan, K.; Yan, X. Iron Oxide Nanozyme: A Multifunctional Enzyme Mimetic for Biomedical Applications. *Theranostics* **2017**, *7*, 3207–3227.

(19) Wu, J.; Wang, X.; Wang, Q.; Lou, Z.; Li, S.; Zhu, Y.; Qin, L.; Wei, H. Nanomaterials with Enzyme-like Characteristics (Nanozymes): Next-Generation Artificial Enzymes (II). *Chem. Soc. Rev.* **2019**, *48*, 1004–1076.

(20) Gao, L.; Zhuang, J.; Nie, L.; Zhang, J.; Zhang, Y.; Gu, N.; Wang, T.; Feng, J.; Yang, D.; Perrett, S.; Yan, X. Intrinsic Peroxidase-Like Activity of Ferromagnetic Nanoparticles. *Nat. Nanotechnol.* **2007**, *2*, 577–583.

(21) Luo, Q.; Li, Y.; Huo, X.; Li, L.; Song, Y.; Chen, S.; Lin, H.; Wang, N. Atomic Chromium Coordinated Graphitic Carbon Nitride for Bioinspired Antibiofouling in Seawater. *Adv. Sci.* **2022**, 2105346.

(22) Yin, W.; Yu, J.; Lv, F.; Yan, L.; Zheng, L. R.; Gu, Z.; Zhao, Y. Functionalized Nano-MoS₂ with Peroxidase Catalytic and Near-Infrared Photothermal Activities for Safe and Synergetic Wound Antibacterial Applications. *ACS Nano* **2016**, *10*, 11000–11011.

(23) Sang, Y.; Li, W.; Liu, H.; Zhang, L.; Wang, H.; Liu, Z.; Ren, J.; Qu, X. Construction of Nanozyme-Hydrogel for Enhanced Capture and Elimination of Bacteria. *Adv. Funct. Mater.* **2019**, *29*, 1900518.

(24) Jiang, D.; Ni, D.; Rosenkrans, Z. T.; Huang, P.; Yan, X.; Cai, W. Nanozyme: New Horizons for Responsive Biomedical Applications. *Chem. Soc. Rev.* **2019**, *48*, 3683–3704.

(25) Herget, K.; Hubach, P.; Pusch, S.; Deglmann, P.; Götz, H.; Gorelik, T. E.; Gural'skiy, I. y. A.; Pfützner, F.; Link, T.; Schenk, S.; Panthöfer, M.; Ksenofontov, V.; Kolb, U.; Opatz, T.; André, R.; Tremel, W. Haloperoxidase Mimicry by CeO_{2-x} Nanorods Combats Biofouling. *Adv. Mater.* **2017**, *29*, 1603823.

(26) Natalio, F.; André, R.; Hartog, A. F.; Stoll, B.; Jochum, K. P.; Wever, R.; Tremel, W. Vanadium Pentoxide Nanoparticles Mimic Vanadium Haloperoxidases and Thwart Biofilm Formation. *Nat. Nanotechnol.* **2012**, *7*, 530–535.

- (27) Ma, L.; Jiang, F.; Fan, X.; Wang, L.; He, C.; Zhou, M.; Li, S.; Luo, H.; Cheng, C.; Qiu, L. Metal-Organic-Framework-Engineered Enzyme-Mimetic Catalysts. *Adv. Mater.* **2020**, *32*, 2003065.
- (28) Ma, Y.; Xie, X.; Yang, W.; Yu, Z.; Sun, X.; Zhang, Y.; Yang, X.; Kimura, H.; Hou, C.; Guo, Z.; Du, W. Recent Advances in Transition Metal Oxides with Different Dimensions as Electrodes for High-Performance Supercapacitors. *Adv. Compos. Hybrid Mater.* **2021**, *4*, 906–924.
- (29) Khan, S. H.; Pathak, B.; Fulekar, M. H. A Study on the Influence of Metal (Fe, Bi, and Ag) Doping on Structural, Optical, and Antimicrobial Activity of ZnO Nanostructures. *Adv. Compos. Hybrid Mater.* **2020**, *3*, 551–569.
- (30) Wang, Y.; Yang, Y.; Wang, L.; Murugadoss, V.; Pan, D.; Xia, H.; Guo, Z. Design, Photoelectric Properties and Electron Transition Mechanism of Cr Doped p-CuGaS₂ Compound Based on Intermediate Band Effect. *Mater. Today Phys.* **2021**, *21*, 100545.
- (31) Jariwala, D.; Sangwan, V. K.; Lauhon, L. J.; Marks, T. J.; Hersam, M. C. Emerging Device Applications for Semiconducting Two-Dimensional Transition Metal Dichalcogenide. *ACS Nano* **2014**, *8*, 1102–1120.
- (32) Liu, M.; Ding, Y.; Wang, L.; Wang, Y.; Wang, X. Band Structure, Photoelectric Properties, and Electronic Transition Mechanism of CuGa_{1-x}Ti_xS₂ Compound Synthesized by Solid-State Reaction Sintering Method. *Adv. Compos. Hybrid Mater.* **2021**, DOI: 10.1007/s42114-021-00315-z.
- (33) Kapadnis, R. S.; Bansode, S. B.; Supekar, A. T.; Bhujbal, P. K.; Kale, S. S.; Jadkar, S. R.; Pathan, H. M. Cadmium Telluride/Cadmium Sulfide Thin Films Solar Cells: A Review. *ES Energy Environ.* **2020**, *10*, 3–12.
- (34) Feng, L.; Zhang, L.; Zhang, S.; Chen, X.; Li, P.; Gao, Y.; Xie, S.; Zhang, A.; Wang, H. Plasma-Assisted Controllable Doping of Nitrogen into MoS₂ Nanosheets as Efficient Nanozymes with Enhanced Peroxidase-Like Catalysis Activity. *ACS Appl. Mater. Interfaces* **2020**, *12*, 17547–17556.
- (35) Wang, L.; Gao, F.; Wang, A.; Chen, X.; Li, H.; Zhang, X.; Zheng, H.; Ji, R.; Li, B.; Yu, X.; Liu, J.; Gu, Z.; Chen, F.; Chen, C. Defect-Rich Adhesive Molybdenum Disulfide/rGO Vertical Heterostructures with Enhanced Nanozyme Activity for Smart Bacterial Killing Application. *Adv. Mater.* **2020**, *32*, 2005423.
- (36) Jaramillo, T. F.; Jørgensen, K. P.; Bonde, J.; Nielsen, J. H.; Hørch, S.; Chorkendorff, I. Identification of Active Edge Sites for Electrochemical H₂ Evolution from MoS₂ Nanocatalysts. *Science* **2007**, *317*, 100–102.
- (37) Zhang, H.; Yu, L.; Chen, T.; Zhou, W.; Lou, X. W. D. Surface Modulation of Hierarchical MoS₂ Nanosheets by Ni Single Atoms for Enhanced Electrocatalytic Hydrogen Evolution. *Adv. Funct. Mater.* **2018**, *28*, 1807086.
- (38) Niu, S.; Cai, J.; Wang, G. Two-dimensional MoS₂ for Hydrogen Evolution Reaction Catalysis: the Electronic Structure Regulation. *Nano Res.* **2020**, *14*, 1985–2002.
- (39) Deng, J.; Li, H.; Wang, S.; Ding, D.; Chen, M.; Liu, C.; Tian, Z.; Novoselov, K. S.; Ma, C.; Deng, D.; Bao, X. Multiscale Structural and Electronic Control of Molybdenum Disulfide Foam for Highly Efficient Hydrogen Production. *Nat. Commun.* **2017**, *8*, 14430.
- (40) Xiao, W.; Liu, P.; Zhang, J.; Song, W.; Feng, Y. P.; Gao, D.; Ding, J. Dual-Functional N Dopants in Edges and Basal Plane of MoS₂ Nanosheets Toward Efficient and Durable Hydrogen Evolution. *Adv. Energy Mater.* **2017**, *7*, 1602086.
- (41) Huang, X.; Leng, M.; Xiao, W.; Li, M.; Ding, J.; Tan, T. L.; Lee, W. S. V.; Xue, J. Activating Basal Planes and S-Terminated Edges of MoS₂ toward More Efficient Hydrogen Evolution. *Adv. Funct. Mater.* **2017**, *27*, 1604943.
- (42) Panthi, G.; Ranjit, R.; Khadka, S.; Gyawali, K. R.; Kim, H.-Y.; Park, M. Characterization and Antibacterial Activity of Rice Grain-Shaped ZnS Nanoparticles Immobilized inside the Polymer Electrospun Nanofibers. *Adv. Compos. Hybrid Mater.* **2020**, *3*, 8–15.
- (43) Hu, P.; Dong, S.; Yuan, F.; Li, X.; Hong, C. Hollow Carbon Microspheres Modified with NiCo₂S₄ Nanosheets as A High-Performance Microwave Absorber. *Adv. Compos. Hybrid Mater.* **2021**, DOI: 10.1007/s42114-021-00318-w.
- (44) Ravi, V. I.; Abhijit, T. S.; Vikram, P. B.; Bikram, P.; Habib, M. P. The Influence of Polysulfide Solvent on the Performance of Cadmium Sulfide Sensitized Zirconium Dioxide-Based Quantum Dots. *ES Mater. Manuf.* **2020**, *9*, 12–20.
- (45) Danish, M.; Qamar, M.; Suliman, M.; Muneer, M. Photoelectrochemical and Photocatalytic Properties of Fe@ZnS QDs/TiO₂ Nanocomposites for Degradation of Different Chromophoric Organic Pollutants in Aqueous Suspension. *Adv. Compos. Hybrid Mater.* **2020**, *3*, 570–582.
- (46) Ganesh, K. R.; Sagar, B. J.; Sachin, R. R.; Yogesh, A. J.; Sunil, V. B.; Avinash, V. R.; Russell, W. C.; Mamta, P. N.; Vijaya, S. J.; Nelson, Y. D.; Sandesh, R. J. Photoelectrochemical Investigation on the Cadmium Sulfide (CdS) Thin Films Prepared using Spin Coating Technique. *ES Mater. Manuf.* **2021**, *11*, 57–64.
- (47) Du, G.; Guo, Z.; Wang, S.; Zeng, R.; Chen, Z.; Liu, H. Superior Stability and High Capacity of Restacked Molybdenum Disulfide as Anode Material for Lithium Ion Batteries. *Chem. Commun.* **2010**, *46*, 1106–1108.
- (48) Jin, Q.; Liu, N.; Dai, C.; Xu, R.; Wu, B.; Yu, G.; Chen, B.; Du, Y. H-2-Directing Strategy on In Situ Synthesis of Co-MoS₂ with Highly Expanded Interlayer for Elegant HER Activity and its Mechanism. *Adv. Energy Mater.* **2020**, *10*, 2000291.
- (49) Huang, J.; He, Y.; Jin, J.; Li, Y.; Dong, Z.; Li, R. A Novel Glucose Sensor Based on MoS₂ Nanosheet Functionalized with Ni Nanoparticles. *Electrochim. Acta* **2014**, *136*, 41–46.
- (50) Pham, K.-C.; Chang, Y.-H.; McPhail, D. S.; Mattevi, C.; Wee, A. T. S.; Chua, D. H. C. Amorphous Molybdenum Sulfide on Graphene-Carbon Nanotube Hybrids as Highly Active Hydrogen Evolution Reaction Catalysts. *ACS Appl. Mater. Interfaces* **2016**, *8*, 5961–5971.
- (51) Xu, Q.; Liu, Y.; Jiang, H.; Hu, Y.; Liu, H.; Li, C. Unsaturated Sulfur Edge Engineering of Strongly Coupled MoS₂ Nanosheet-Carbon Macroporous Hybrid Catalyst for Enhanced Hydrogen Generation. *Adv. Energy Mater.* **2019**, *9*, 1802553.
- (52) Chen, P.; Zhou, T.; Chen, M.; Tong, Y.; Zhang, N.; Peng, X.; Chu, W.; Wu, X.; Wu, C.; Xie, Y. Enhanced Catalytic Activity in Nitrogen-Anion Modified Metallic Cobalt Disulfide Porous Nanowire Arrays for Hydrogen Evolution. *ACS Catal.* **2017**, *7*, 7405–7411.
- (53) Ni, Q.; Bai, Y.; Guo, S.; Ren, H.; Chen, G.; Wang, Z.; Wu, F.; Wu, C. Carbon Nanofiber Elastically Confined Nanoflowers: A Highly Efficient Design for Molybdenum Disulfide-Based Flexible Anodes Toward Fast Sodium Storage. *ACS Appl. Mater. Interfaces* **2019**, *11*, 5183–5192.
- (54) Sun, T.; Wang, J.; Chi, X.; Lin, Y.; Chen, Z.; Ling, X.; Qiu, C.; Xu, Y.; Song, L.; Chen, W.; Su, C. Engineering the Electronic Structure of MoS₂ Nanorods by N and Mn Dopants for Ultra-Efficient Hydrogen Production. *ACS Catal.* **2018**, *8*, 7585–7592.
- (55) Liu, G.; Robertson, A. W.; Li, M. M.-J.; Kuo, W. C. H.; Darby, M. T.; Muhieddine, M. H.; Lin, Y.-C.; Suenaga, K.; Stamatakis, M.; Warner, J. H.; Tsang, S. C. E. MoS₂ Monolayer Catalyst Doped with Isolated Co Atoms for the Hydrodeoxygenation Reaction. *Nat. Chem.* **2017**, *9*, 810–816.
- (56) Shi, Y.; Zhou, Y.; Yang, D.-R.; Xu, W.-X.; Wang, C.; Wang, F.-B.; Xu, J.-J.; Xia, X.-H.; Chen, H.-Y. Energy Level Engineering of MoS₂ by Transition-Metal Doping for Accelerating Hydrogen Evolution Reaction. *J. Am. Chem. Soc.* **2017**, *139*, 15479–15485.
- (57) Tran, P. D.; Tran, T. V.; Orio, M.; Torelli, S.; Truong, Q. D.; Nayuki, K.; Sasaki, Y.; Chiam, S. Y.; Yi, R.; Honma, I.; Barber, J.; Artero, V. Coordination Polymer Structure and Revisited Hydrogen Evolution Catalytic Mechanism for Amorphous Molybdenum Sulfide. *Nat. Mater.* **2016**, *15*, 640–646.
- (58) Lei, F.; Sun, Y.; Liu, K.; Gao, S.; Liang, L.; Pan, B.; Xie, Y. Oxygen Vacancies Confined in Ultrathin Indium Oxide Porous Sheets for Promoted Visible-Light Water Splitting. *J. Am. Chem. Soc.* **2014**, *136*, 6826–6829.
- (59) Busetto, L.; Vaccari, A.; Martini, G. Electron Spin Resonance of Paramagnetic Species as a Tool for Studying the Thermal

Decomposition of Molybdenum Trisulfide. *J. Phys. Chem.* **1981**, *85*, 1927–1930.

(60) Luo, Z.; Wang, Y.; Wang, X.; Jin, Z.; Wu, Z.; Ge, J.; Liu, C.; Xing, W. Simultaneously Engineering Electron Conductivity Site Density and Intrinsic Activity of MoS₂ via the Cation and Anion Codoping Strategy. *ACS Appl. Mater. Interfaces* **2019**, *11*, 39782–39788.

(61) Colpas, G. J.; Hamstra, B. J.; Kampf, J. W.; Pecoraro, V. L. Functional Models for Vanadium Haloperoxidase: Reactivity and Mechanism of Halide Oxidation. *J. Am. Chem. Soc.* **1996**, *118*, 3469–3478.

(62) Sokolov, A. V.; Kostevich, V. A.; Kozlov, S. O.; Donskyi, I. S.; Vlasova, I. I.; Rudenko, A. O.; Zakharova, E. T.; Vasilyev, V. B.; Panasenko, O. M. Kinetic Method for Assaying the Halogenating Activity of Myeloperoxidase Based on Reaction of Celestine Blue B with Taurine Halogenamines. *Free Radical Res.* **2015**, *49*, 777–789.

(63) Flemming, H.-C.; Schaule, G. Biofouling on Membranes—A Microbiological Approach. *Desalination* **1988**, *70*, 95–119.

(64) Bendinger, B.; Rijnaarts, H. H. M.; Altendorf, L.; Zehnder, A. J. B. Physicochemical Cell Surface and Adhesive Properties of Coryneform Bacteria Related to the Presence and Chain Length of Mycolic Acids. *Appl. Environ. Microbiol.* **1993**, *59*, 3937–3977.

(65) Danielle, R. S.; Satish, C. B. M. Impact of Bromide Exposure on Natural Organochlorine Loss from Coastal Wetland Soils in the Winyah Bay, South Carolina. *Environ. Sci.: Processes Impacts* **2020**, *22*, 642–652.

(66) Hopwood, M. J.; Rapp, I.; Schlosser, C.; Achterberg, E. P. Hydrogen Peroxide in Deep Waters from the Mediterranean Sea, South Atlantic and South Pacific Oceans. *Sci. Rep.* **2017**, *7*, 43436.



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